

3. (a) An organisation has bought a software package containing several applications to be used in the various departments of the organisation. (1 mark)
- (i) State the name of this type of software package. (2 marks)
 - (ii) Give ONE advantage and ONE disadvantage of using this type of software package.
- (b) State the type of processing that is BEST suited for EACH of the following tasks:
- (i) Processing of monthly electricity bills (1 mark)
 - (ii) Processing of airline reservations (1 mark)
- (c) (i) Convert Octal 243 to decimal, binary and hexadecimal. (3 marks)
- (ii) Convert -856 to Binary Coded Decimal (BCD). (2 marks)
- Total 10 marks**

3. (a) Integrated Software Package

(i) Advantage - Since a number of different application software is included in this package, it is not very costly when compared to buying one software at a time.

• Disadvantage - Some programs cannot be opened in these software programs. There is a limit of application software & programs available to the user.

(b) (i) Batch processing

(ii) Real-time processing

Write the question number in this column.

(c) (i)	2	243	
	2	121	R ₁ ↑
	2	60	R ₁
	2	30	R ₀
	2	15	R ₁
	2	7	R ₁
	2	3	R ₁
	2	1	R ₁
	0		R ₁

$$243 = 11111011$$

(ii) -856

$$\text{BCD} = 1111\ 1000\ 0101\ 0110$$

$$\begin{array}{r} 2 \quad 8 \\ 2 \quad 4 \quad R_0 \\ 2 \quad 2 \quad R_0 \\ 2 \quad 2 \quad R_0 \\ 0 \quad R_1 \end{array}$$

$$8_{10} = 1009$$

Write the question number in this column.

CSEC Information Technology

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Comments

Question 3 (a)

The question tested the candidate's ability to distinguish between systems and application programs and the advantages and disadvantages of the use of various types of application programs.

For Part (i), the candidate stated the correct type of software application used in the scenario given.

In Part (ii), the candidate demonstrated a good understanding of the advantages and disadvantages of the software package given in the scenario.

Question 3 (b)

For both Parts (i) and (ii), the candidate stated the correct processing used in the scenarios given.

This candidate's responses represented an excellent performance on this section of the question for which full marks were awarded.

Question 3 (c)

The candidate did not complete Part (i) in its entirety and did not record the correct answer for the part attempted.

In Part (ii), the candidate demonstrated the ability to correctly convert to BCD.

Summary

The candidate demonstrated good knowledge and understanding of system and application programs and processing modes. However, there was no evidence that the candidate could manipulate binary, decimal and hexadecimal systems using conversion.